

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil Engineering) SEMESTER - 7 Winter 2025 (Supply.)

Course :Bachelor of Technology (Civil Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 7

Subject Code & Name: BTCVE705F - ADVANCED RC DESIGN

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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Q1. Objective type questions. (Compulsory Question)

12

- 1 **A slab is a structural element that primarily resists loads in:**
a) One direction b) Two directions c) Three directions d) None
- 2 **For a circular slab simply supported at the edges and subjected to a uniformly distributed load (UDL), the bending moment at the center is:**
A. Maximum and positive B. Zero
C. Maximum and negative D. Constant throughout the slab
- 3 **As per IS 456:2000, for a flat slab, the minimum thickness of the slab (excluding the drop panel) should not be less than:**
A. 150 mm B. 100 mm C. 175 mm D. 125 mm
- 4 **In the analysis of flat slabs, the effective length of a column capital is usually taken considering a maximum flare angle with the vertical of:**
A. 30^0 B. 45^0 C. 60^0 D. 90^0
- 5 **In a flat slab, the maximum spacing of the main reinforcement bars (for moment resistance) shall not exceed:**
A. 300 mm or 5d, whichever is less.
B. 300 mm or 3d, whichever is less.
C. 450 mm or 4d, whichever is less.
D. 300 mm or 4d, whichever is less.
- 6 **In flat slab construction, openings are preferably located:**
A. Near column heads B. In the drop region
C. Away from column heads and slab corners D. Anywhere in the slab
- 7 **A dome is a structural element that primarily resists:**
a) Bending stresses b) Axial and hoop stresses
c) Shear stresses d) Torsional stresses
- 8 **The stress resulting from wind load on a dome is generally:**
A. Only compressive B. Only tensile C. Bending stress
D. Both tensile and compressive, depending on direction of load
- 9 **The minimum thickness of an RCC spherical dome is generally:**
a) 50 mm b) 100 mm c) 150 mm d) 200 mm
- 10 **The primary function of a silo is to:**
a) Store liquid materials b) Store granular materials
c) Act as a support structure for bridges d) Store water
- 11 **The shape of a hopper bottom is generally:**
a) Flat b) Conical or pyramidal c) Cylindrical d) Rectangular
- 12 **The height of a chimney is primarily determined by:**
a) The aesthetic requirement b) Amount & type of pollutants emitted
c) The weight of the structure d) The wind velocity only

- Q2. Solve the following.
- A) Explain the behavior of circular slabs under uniformly distributed load (UDL) when Fixed at the edges. 6
 - B) Design a circular slab of 3.5 m diameter to cover an underground sump. The slab is simply supported at the periphery by a wall 200 mm thick. Assume a finish load of 1.0 kN/m^2 and live loads of 4.0 kN/m^2 . Use M 20 concrete and Fe 415 steel. Assume *mild exposure* conditions. 6
- Q3. Solve the following.
- A) Explain the merits of flat slab construction over conventional beam-slab systems. 6
 - B) Explain how the 'unbalanced' moment is transferred from the slab to the column in flat slabs. 6
- Q4. Solve Any Two of the following.
- A) Explain the nature of stresses in spherical domes. How do these stresses influence the structural behavior of domes? 6
 - B) A reinforced concrete (RC) spherical dome has a radius of 6 m and is subjected to a uniformly distributed load of 4 kN/m^2 including self-weight. The thickness of the dome is 150 mm. Determine the *hoop stress* at a latitude where the angle $\theta = 30^\circ$ 6
 - C) Explain the design considerations for reinforced concrete domes. 6
- Q5. Solve Any Two of the following.
- A) Explain the difference between a bunker and a silo. 6
 - B) Describe the failure modes in bunkers and silos due to material pressure and loading conditions. 6
 - C) Write a short note on the failure modes of silos and bunkers. 6
- Q6. Solve Any Two of the following.
- A) Describe the effect of wind pressure on tall chimneys. How is wind-induced stress calculated as per IS codes? 6
 - B) Derive the expression for stresses developed in the chimney shaft due to self-weight and wind pressure. 6
 - C) Explain effect of wind shear on horizontal reinforcement in tall chimneys. 6

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